

KYZYLZHAR, Kazakhstan

WMO: 355760

Lat: 48.300N

Lon: 69.650E

Elev: 1184

StdP: 14.08

Time Zone: 6.00 (E06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-19.3	-30.8	1.0	-23.6	-25.6	1.4	-19.0	27.9	18.1	23.4	20.0	4.5	50	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.4	94.9	65.4	91.9	64.1	88.9	63.3	67.5	87.5	66.1	86.4	64.8	84.5	7.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.3	84.7	72.8	59.3	78.8	73.3	57.1	72.6	73.3	32.7	87.2	31.5	86.4	30.5	84.3	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.2	16.4	DB	-27.7	100.3	5.9	2.8	-32.0	102.3	-35.4	103.9	-38.7	105.4	-43.0	107.4	(4)
(5)			WB	-28.0	69.9	5.8	2.4	-32.2	71.6	-35.6	72.9	-38.9	74.3	-43.1	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.8	4.1	6.3	25.6	48.6	60.2	70.3	73.3	70.7	58.6	43.5	26.8	11.8
	DBStd	26.79	12.83	13.24	13.22	9.22	7.64	6.83	5.96	6.80	8.41	8.98	11.98	13.34
	HDD50	5681	1424	1222	759	132	10	0	0	0	17	235	697	1185
	HDD65	9183	1889	1642	1221	495	185	28	7	25	229	666	1146	1650
	CDD50	2704	0	0	3	89	327	610	723	642	274	35	1	0
	CDD65	730	0	0	0	2	37	188	264	202	37	0	0	0
	CDH74	9741	0	0	1	63	686	2547	3336	2535	561	12	0	0
	CDH80	4186	0	0	0	10	179	1123	1554	1109	211	0	0	0
Wind	WSAvg	6.7	6.5	6.9	7.9	7.6	7.3	6.9	6.4	6.2	5.7	6.0	6.2	6.7
Precipitation	PrecAvg	7.6	0.7	0.5	0.5	0.7	0.9	0.6	0.8	0.5	0.4	0.6	0.6	0.6
	PrecMax	12.8	2.8	1.3	1.5	1.9	2.7	3.1	3.3	2.1	1.5	2.0	2.0	1.5
	PrecMin	3.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.2
	PrecStd	2.0	0.5	0.3	0.4	0.6	0.7	0.6	0.7	0.5	0.4	0.5	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	35.5	63.4	80.8	89.1	97.5	99.4	97.4	93.0	75.5	57.1	36.4
		MCWB	31.6	32.5	48.6	57.8	62.6	66.9	67.6	66.0	63.7	54.5	49.4	32.5
	2%	DB	31.3	33.0	54.8	74.6	84.8	93.0	95.5	93.3	86.3	68.8	50.5	33.9
		MCWB	29.3	31.1	45.6	54.9	60.4	64.5	65.9	64.2	60.9	51.3	45.8	31.4
	5%	DB	27.7	30.3	47.4	69.9	81.3	90.0	92.3	89.9	80.6	63.5	45.4	31.8
		MCWB	25.9	28.6	41.3	53.2	58.9	63.5	64.8	63.1	58.4	49.0	40.4	29.6
	10%	DB	23.7	27.3	40.9	64.9	77.9	86.7	88.6	86.5	75.4	58.2	40.9	29.0
		MCWB	22.2	25.7	36.8	51.0	57.5	62.5	64.0	62.5	55.9	46.6	37.1	27.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.7	33.0	50.3	60.0	65.9	69.2	70.4	67.3	64.2	55.5	51.2	33.4
		MCDB	33.2	34.6	61.3	74.2	86.3	91.1	92.2	88.6	88.3	71.3	55.3	35.1
	2%	WB	29.4	31.2	45.6	56.9	62.2	66.7	68.2	65.9	61.7	52.2	46.0	31.5
		MCDB	31.3	33.0	54.2	72.0	80.5	86.7	87.4	88.8	85.2	65.2	49.9	33.4
	5%	WB	26.0	28.7	41.3	54.5	60.3	65.1	66.6	64.7	58.9	49.8	40.9	29.7
		MCDB	27.8	30.3	47.3	67.2	77.6	85.0	85.9	86.0	78.7	61.4	44.3	31.8
	10%	WB	22.2	25.7	37.0	52.0	58.6	63.7	65.4	63.2	56.6	47.6	37.1	27.2
		MCDB	23.6	27.2	40.8	63.4	74.7	83.4	84.2	83.7	73.9	57.5	40.4	29.2
Mean Daily Temperature Range	5% DB	MDBR	15.5	17.8	16.7	22.2	25.9	25.0	24.4	24.6	24.2	21.1	14.7	14.5
		MCDBR	11.2	13.6	22.8	30.1	31.1	29.7	29.5	29.1	30.7	29.5	18.8	12.8
		MCWBR	10.5	12.9	14.9	14.5	13.3	10.6	10.0	10.7	13.1	16.4	13.6	11.2
	5% WB	MCDBR	11.3	14.1	22.0	26.0	27.6	26.3	26.0	27.2	29.0	26.3	17.0	12.4
		MCWBR	10.8	13.5	14.7	14.5	13.0	10.8	10.3	11.3	13.4	15.8	13.1	11.2
Clear-Sky Solar Irradiance	taub	0.305	0.340	0.390	0.419	0.428	0.422	0.429	0.395	0.352	0.347	0.336	0.311	
	taud	2.066	1.985	2.042	2.092	2.124	2.212	2.207	2.284	2.357	2.279	2.183	2.119	
	Ebn at Noon	229	247	256	262	265	267	263	268	270	247	218	210	
	Edh at Noon	29	40	44	47	47	44	43	38	32	30	26	24	
All-Sky Solar Radiation	RadAvg	517	893	1268	1644	2087	2212	2081	1895	1462	860	444	362	
	RadStd	80	65	121	150	133	126	108	123	71	86	44	55	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page